

Algebraic Topology I: Homework 3

Derived Functors

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. An object P is *projective* if $\text{Mor}_{\mathcal{A}}(P, -)$ takes epimorphisms to surjections; an object I is *injective* if $\text{Mor}_{\mathcal{A}}(-, I)$ takes monomorphisms to surjections. The category $\mathcal{A}$ has *enough projectives* if every object is a quotient of a projective object, and *enough injectives* if every object embeds in an injective object.

Use homological grading for projective resolutions and cohomological grading for injective resolutions. All functors between abelian categories below are additive. You may use Homework 2, including homotopy invariance of homology and the long exact sequence of a short exact sequence of complexes, as well as their cochain analogues. In an abstract abelian category, use morphisms and universal properties; element arguments are permitted for modules.

Exercise 1. Resolutions and Comparison Maps

Assume that $\mathcal{A}$ has enough projectives. A *projective resolution* of M is an exact augmented complex

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} M \longrightarrow 0$$

with every P_n projective.

- (a) Construct a projective resolution of every $M \in \mathcal{A}$ by successively resolving kernels. For modules over a unital ring R , explain why free modules suffice. Write an arbitrary free module as $\bigoplus_{i \in I} R$.
- (b) Given projective resolutions $P_{\bullet} \rightarrow M$ and $Q_{\bullet} \rightarrow N$, prove that every $f : M \rightarrow N$ lifts to a chain map $\tilde{f} : P_{\bullet} \rightarrow Q_{\bullet}$. Prove that any two lifts of the same f are chain homotopic. Specify the induction that constructs both the lift and the homotopy.
- (c) Deduce that two projective resolutions of the same object are chain homotopy equivalent by maps lifting its identity. Explain why the induced maps on homology after applying any additive functor are independent of all choices and compatible with composition.

Exercise 2. Left Derived Functors

Assume that $\mathcal{A}$ has enough projectives, and let $F : \mathcal{A} \rightarrow \mathcal{B}$ be right exact. For a projective resolution $P_{\bullet} \rightarrow M$, set

$$L_n F(M) = H_n(F(P_{\bullet})), \quad n \geq 0.$$

- (a) Using Exercise 1, define $L_n F$ on morphisms and prove that it is an additive functor, well defined up to canonical natural isomorphism independently of the chosen resolutions.
- (b) Construct a natural isomorphism $L_0 F \cong F$. Identify precisely where right exactness is used.
- (c) Prove that $L_n F(P) = 0$ for every projective P and $n > 0$. If F is exact, prove that $L_n F(M) = 0$ for every M and $n > 0$.

Exercise 3. The Horseshoe Lemma and Connecting Maps

Assume that $\mathcal{A}$ has enough projectives. Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be exact, and choose projective resolutions $P'_\bullet \rightarrow M'$ and $P''_\bullet \rightarrow M''$.

- (a) Construct a projective resolution $P_\bullet \rightarrow M$ and a short exact sequence of augmented complexes lifting the given sequence such that

$$0 \longrightarrow P'_n \longrightarrow P_n \longrightarrow P''_n \longrightarrow 0$$

is split exact for every $n \geq 0$. You may take $P_n = P'_n \oplus P''_n$; construct its differential and augmentation. The given sequence of objects is not assumed to split.

- (b) For a right exact $F : \mathcal{A} \rightarrow \mathcal{B}$, explain why applying F gives a short exact sequence of the unaugmented complexes, even though F need not be left exact. Deduce a long exact sequence ending in

$$\cdots \rightarrow L_1 F(M) \rightarrow L_1 F(M'') \xrightarrow{\delta} F(M') \rightarrow F(M) \rightarrow F(M'') \rightarrow 0.$$

Describe the connecting maps in all positive degrees.

- (c) Prove that this long exact sequence is independent of the horseshoe choices and natural in morphisms of short exact sequences. Explain how to choose comparison maps compatible with the inclusions and projections.

Exercise 4. Right Derived Functors

Now assume that $\mathcal{A}$ has enough injectives and let $G : \mathcal{A} \rightarrow \mathcal{B}$ be left exact. An *injective resolution* of M is an exact sequence $0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow \cdots$ with each I^n injective. Set

$$R^n G(M) = H^n(G(I^\bullet)), \quad n \geq 0.$$

- (a) State and prove the injective analogues of the existence and comparison results in Exercise 1. Give the direction of the comparison map associated to $f : M \rightarrow N$. Deduce that $R^n G$ is a well-defined additive functor up to canonical natural isomorphism.
- (b) Prove that $R^0 G \cong G$, and that $R^n G(I) = 0$ for injective I and $n > 0$. Construct the long exact sequence beginning in

$$0 \rightarrow G(M') \rightarrow G(M) \rightarrow G(M'') \xrightarrow{\delta} R^1 G(M') \rightarrow R^1 G(M) \rightarrow R^1 G(M'') \rightarrow \cdots.$$

You may obtain the injective horseshoe lemma by duality.

- (c) Show that G is exact if and only if $R^1 G(M) = 0$ for every M . Prove the corresponding criterion for a right exact F in terms of $L_1 F$, assuming enough projectives.

Exercise 5. Dimension Shifting and Acyclic Resolutions

Continue with F as in Exercise 2 and G as in Exercise 4, making the corresponding assumptions on $\mathcal{A}$. Call A F -acyclic if $L_n F(A) = 0$ for every $n > 0$.

- (a) If $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$ is exact and P is projective, construct natural isomorphisms

$$L_{n+1}F(M) \cong L_n F(K) \quad (n \geq 1)$$

and the exact sequence $0 \rightarrow L_1 F(M) \rightarrow F(K) \rightarrow F(P) \rightarrow F(M) \rightarrow 0$. Explain why the displayed isomorphism is not asserted for $n = 0$.

- (b) Give and prove the dual statements for an exact sequence $0 \rightarrow M \rightarrow I \rightarrow C \rightarrow 0$ with I injective.
- (c) Suppose $\cdots \rightarrow A_1 \rightarrow A_0 \rightarrow M \rightarrow 0$ is an exact resolution by F -acyclic objects. Prove that $H_n(F(A_\bullet)) \cong L_n F(M)$ for every $n \geq 0$. Use the short exact sequences involving its successive kernels and dimension shifting. No finite-length assumption is made.

Exercise 6. Tensor Products and Tor

Let R be a commutative unital ring and N an R -module. Define

$$\mathrm{Tor}_n^R(M, N) = L_n(- \otimes_R N)(M).$$

An R -module N is *flat* if $- \otimes_R N$ is exact.

- (a) Check that $- \otimes_R N$ is right exact. Use Exercise 4 to prove that N is flat if and only if $\mathrm{Tor}_1^R(M, N) = 0$ for every M .
- (b) Let $r \in R$ be a non-zero-divisor. Use the resolution $0 \rightarrow R \xrightarrow{r} R \rightarrow R/(r) \rightarrow 0$ to compute $\mathrm{Tor}_n^R(R/(r), N)$ for all $n \geq 0$. Express the answer using N/rN and $N[r] = \{x \in N : rx = 0\}$. Explain why the same resolution cannot be used when r is a zero-divisor.
- (c) Take $R = \mathbb{Z}$ and positive integers $m, n \geq 2$. Compute $\mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/m, \mathbb{Z}/n)$ for every $i \geq 0$. Identify the connecting injection $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/m, N) \rightarrow N$ obtained by tensoring $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}/m \rightarrow 0$.

Exercise 7. Ext and Extensions

Let R be a unital ring, not necessarily commutative, and work with left R -modules. Fix a module M and define

$$\mathrm{Ext}_R^n(M, N) = R^n \mathrm{Mor}_R(M, -)(N).$$

You may use that module categories have enough injectives and the standard comparison theorem identifying these groups with $H^n(\mathrm{Mor}_R(P_\bullet, N))$ for a projective resolution $P_\bullet \rightarrow M$. Here the cochain differential sends $u : P_n \rightarrow N$ to $u \circ d_{n+1}$. For a noncommutative R , these are abelian groups, not in general left R -modules.

- (a) Given $0 \rightarrow K \xrightarrow{j} P \rightarrow M \rightarrow 0$ with P projective, prove that

$$\mathrm{Ext}_R^1(M, N) \cong \mathrm{coker}(\mathrm{Mor}_R(P, N) \xrightarrow{j^*} \mathrm{Mor}_R(K, N)).$$

For $u : K \rightarrow N$, form the pushout along u and obtain an extension $0 \rightarrow N \rightarrow E_u \rightarrow M \rightarrow 0$. Prove that the resulting construction identifies the displayed cokernel with equivalence classes of extensions, where equivalences induce the identities on N and M . Show that the zero class corresponds exactly to split extensions.

- (b) Deduce that M is projective if and only if $\mathrm{Ext}_R^1(M, N) = 0$ for every left R -module N .
- (c) For $R = \mathbb{Z}$, $m \geq 2$, and an abelian group N , compute $\mathrm{Ext}_{\mathbb{Z}}^i(\mathbb{Z}/m, N)$ in every degree. Then take $N = \mathbb{Z}/n$ and express the answer in terms of $\gcd(m, n)$.

Exercise 8. An Infinite Resolution

Let k be a field, let $R = k[\varepsilon]/(\varepsilon^2)$, and regard $k = R/(\varepsilon)$ as an R -module.

- (a) Prove that the augmented complex

$$\cdots \xrightarrow{\varepsilon} R \xrightarrow{\varepsilon} R \xrightarrow{\varepsilon} R \longrightarrow k \longrightarrow 0$$

is a free resolution of k by computing its kernels and images.

- (b) Compute $\mathrm{Tor}_n^R(k, k)$ and $\mathrm{Ext}_R^n(k, k)$ for every $n \geq 0$, including the differentials in the complexes used for these computations.
- (c) Prove that k has no projective resolution of finite length over R . Contrast this with the computations over $\mathbb{Z}$ in Exercises 6 and 7.